Name: _____

TOPIC TEST 2

Chapter 6: Trigonometric functions

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks)
- Part B: 7 short answer questions (42 marks)
- Part C: 3 analysis questions (27 marks)

Part A

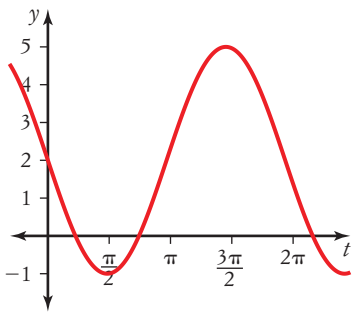
5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 Find the amplitude and period of the function $f(x) = 2 - 5\cos(2\pi x)$.

A Amplitude 2, period 2π
B Amplitude 5, period π
C Amplitude -5, period 1
D Amplitude -5, period π
E Amplitude 5, period 1

[1 mark]

2 The equation of the graph below is:



A $y = 3\sin(x) + 2$
B $y = 3 - 2\sin(x)$
C $y = 2 - 3\sin(x)$
D $y = 2 + 6\sin(x)$
E $y = 2 - 6\sin(x)$

[1 mark]

3 Which statement is not correct?

A $\sin^2(x) + \cos^2(x) = 1$

B $\tan(x) = \dfrac{\cos(x)}{\sin(x)}$

C $\cos(x - y) = \cos(x)\cos(y) + \sin(x)\sin(y)$

D $\cos\left(x - \dfrac{\pi}{2}\right) = \sin(x)$

E $\sin\left(\dfrac{\pi}{2} - x\right) = \cos(x)$

[1 mark]

4 Solve $\cos(2x) = -1$ for $0 \leq x \leq 2\pi$.

A $x = \dfrac{\pi}{2}, \dfrac{3\pi}{2}$

B $x = \left(\dfrac{\pi}{2}\right)$

C $x = 0, \pi, 2\pi$

D $x = \left(\dfrac{3\pi}{2}\right)$

E $x = \pi, \left(\dfrac{3\pi}{2}\right)$

[1 mark]

5 Find the exact value of $\sin(105°)$.

A $\dfrac{\sqrt{3} - 1}{2\sqrt{2}}$

B $\dfrac{1 - \sqrt{3}}{2\sqrt{2}}$

C $\dfrac{\sqrt{3} + 1}{\sqrt{2}}$

D $\dfrac{\sqrt{3} - 1}{\sqrt{2}}$

E $\dfrac{\sqrt{3} + 1}{2\sqrt{2}}$

[1 mark]

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Part B

7 short answer questions
42 marks
Show your working where appropriate.

1 Find the value of:

a $\cos(0)$ _____

b $\tan(2\pi)$ _____

c $\sin\left(\frac{3\pi}{2}\right)$ _____

d $\cos(\pi)$ _____

e $\tan(180^\circ)$ _____

[5 marks]

2 For the domain $0 \leq x \leq 2\pi$, sketch the graph of:

a $y = \tan(x)$

b $y = \sin(x) + 3$

c $y = 1 - 2\cos(x)$

d $y = 2\cos\left(\frac{x}{2}\right)$

e $y = 1 + \tan\left(x + \frac{\pi}{4}\right)$

[10 marks]

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3 Expand

a $\tan(2a + b)$

b $\cos(5a - 3b)$

c $\sin(7p - q)$

[3 marks]

4 By drawing the graphs of $y = 3\cos(2x)$ and $y = x + 1$ on the same set of axes, graphically solve $3\cos(2x) = x + 1$. You may use a CAS calculator.

[4 marks]

5 For $0° \leq x \leq 360°$ solve

a $\tan(x) = 1$

b $\sin(x) = -\frac{\sqrt{3}}{2}$

c $\tan(2x) = \frac{1}{\sqrt{3}}$

[11 marks]

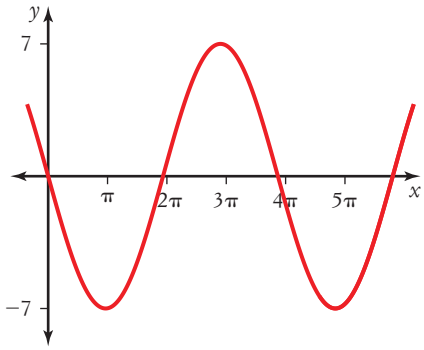
6 Solve $2\cos^2(\theta) - 1 = 0$ for $0 \leq \theta \leq 2\pi$.

[5 marks]

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7 **a** State the amplitude and period of the function below.

b Find an equation that models this function.



[4 marks]

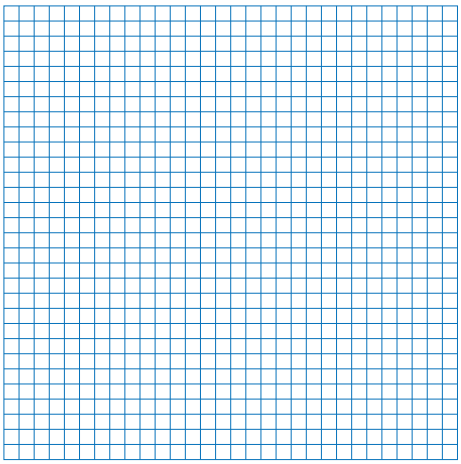
Part C

3 analysis questions
27 marks
Show your working where appropriate.

1 Below is a table showing the heights of the tides over a three day period.

Day	Friday				Saturday				Sunday			
Time	6.20 a.m.	11.55 a.m.	6.15 p.m.	11.48 p.m.	6.20 a.m.	11.55 a.m.	6.15 p.m.	11.48 p.m.	6.20 a.m.	11.55 a.m.	6.15 p.m.	11.48 p.m.
Tide (m)	3.2	1.1	3.4	1.3	3.2	1.2	3.5	1.1	3.4	1.2	3.5	1.3

a Draw a graph showing the tides.



b Find the period and amplitude.

c Estimate the height of the tide at around 8 a.m. on Friday.

[6 marks]

2 In an unusual meteorological investigation, the temperature, T °C, in a town in central Queensland was found to fluctuate approximately according to the rule

$T = 25 + 6 \sin(0.1\pi t)$, where t is the number of hours after 10:00 a.m.

a Sketch a graph of the temperature fluctuations for a sufficient number of hours such that you are able to determine the maximum and minimum temperatures for that day and the next night.

b Use the graph to determine the maximum and minimum temperatures.

c When did they occur?

d At what time was the temperature:

i 27°C? _____

ii 20°C? _____

e Why was this unusual?

[9 marks]

3 **a** Show how $\cos(2x)$ can be expressed in terms of $\cos(x)$.

b Show how sin $(2x)$ can be expressed in terms of sin (x) and cos (x).

c Hence show how cos $(3x)$ can be expressed in terms of cos (x).

[12 marks]
Total marks: 74

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Answers

Part A **[1 mark each]**

1 Amplitude $= 5$

Period $= \dfrac{2\pi}{b} = \dfrac{2\pi}{2\pi} = 1$

$\therefore$ **E** **[1 mark]**

2 Amplitude $= 3$

Period $= 2\pi$

$y = \sin(x)$ is reflected in the x-axis and translated up 2 units.

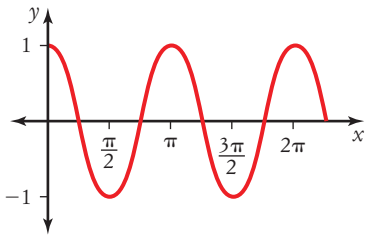
$y = 2 - 3\sin(x)$

$\therefore$ **C** **[1 mark]**

3 $\tan(x) = \dfrac{\sin(x)}{\cos(x)}$

$\therefore$ **B** **[1 mark]**

4 $y = \cos(2x)$ has period $\dfrac{2\pi}{2} = \pi$



From the graph $x = \dfrac{\pi}{2}, \dfrac{3\pi}{2}$

$\therefore$ **A** **[1 mark]**

5 $\sin(105°) = \sin(60° + 45°)$

$= \sin(60°)\cos(45°) + \cos(60°)\sin(45°)$

$= \dfrac{\sqrt{3}}{2} \times \dfrac{1}{\sqrt{2}} + \dfrac{1}{2} \times \dfrac{1}{\sqrt{2}}$

$= \dfrac{\sqrt{3}}{2\sqrt{2}} + \dfrac{1}{2\sqrt{2}}$

$= \dfrac{\sqrt{3}+1}{2\sqrt{2}}$

$\therefore$ **E** **[1 mark]**

Part B

1 a $\cos(0) = 1$ **[1 mark]**

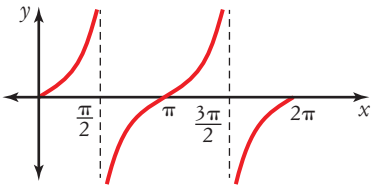
b $\tan(2\pi) = 0$ **[1 mark]**

c $\sin\left(\frac{3\pi}{2}\right) = -1$ **[1 mark]**

d $\cos(\pi) = -1$ **[1 mark]**

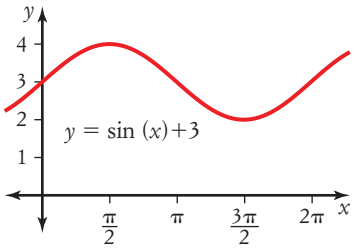
e $\tan(180°) = 0$ **[1 mark]**

2 a Period $= \pi$ **[1 mark]**



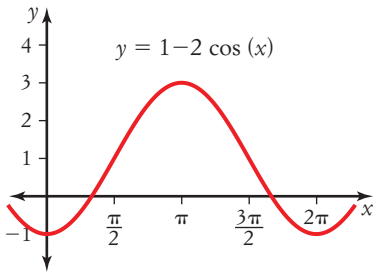
[1 mark]

b $y = \sin(x)$ translated 3 units up. **[1 mark]**



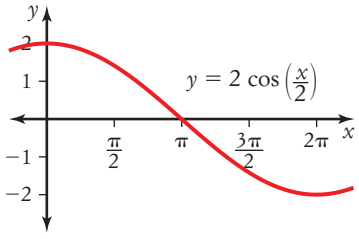
[1 mark]

c Amplitude $= 2$, translated 1 unit up, reflected in the x-axis. **[1 mark]**



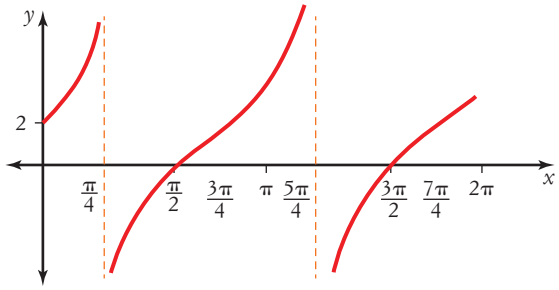
[1 mark]

d Period $= \frac{2\pi}{\frac{1}{2}} = 4\pi$, amplitude $= 2$ **[1 mark]**



[1 mark]

e The graph $y = \tan(x)$ is translated $\frac{\pi}{4}$ units to the left and up 1 unit. **[1 mark]**



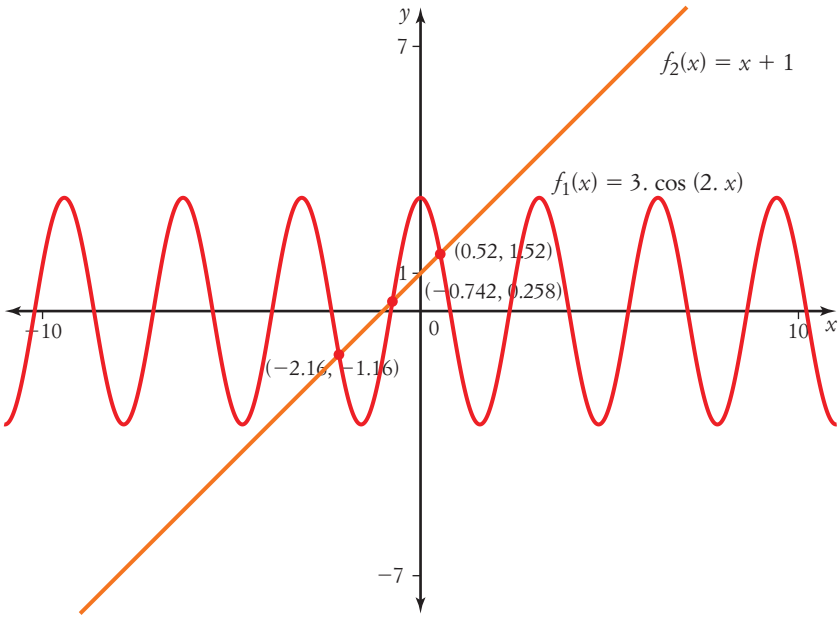
[1 mark]

3 **a** $\tan(2a + b) = \dfrac{\tan(2a) + \tan(b)}{1 - \tan(2a)\tan(b)}$ **[1 mark]**

b $\cos(5a - 3b) = \cos(5a)\cos(3b) + \sin(5a)\sin(3b)$ **[1 mark]**

c $\sin(7p - q) = \sin(7p)\cos(q) - \cos(7p)\sin(q)$ **[1 mark]**

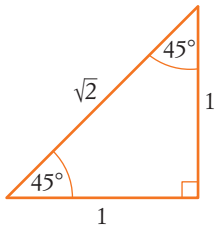
4 On the CAS calculator or freehand, draw both graphs.



[3 marks]

$x = -2.16, -0.742, 0.52$ **[1 mark]**

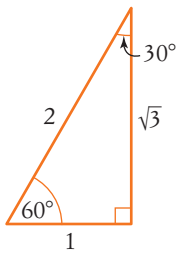
5 **a** From the triangle, $\tan(45°) = 1$ **[1 mark]**



$x = 45°, 180° + 45°$ (tan is positive in the 1st and 3rd quadrants) **[1 mark]**

$= 45°, 225°$ **[1 mark]**

b From the triangle, $\sin(60°) = \frac{\sqrt{3}}{2}$ **[1 mark]**



$x = 180° + 60°, 360° - 60°$ (sin is negative in the 3rd and 4th quadrants) **[1 mark]**

$= 240°, 300°$ **[1 mark]**

c From the triangle, $\tan(30°) = \frac{1}{\sqrt{3}}$ **[1 mark]**

$0° \leq x \leq 360°$

$0° \leq 2x \leq 720°$ **[1 mark]**

$2x = 30°, 180° + 30°, 360° + 30°, 360° + 180° + 30°$ (tan is positive in the 1st and 3rd quadrants) **[1 mark]**

$= 30°, 210°, 390°, 570°$ **[1 mark]**

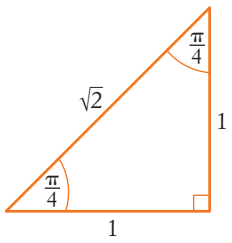
$x = 15°, 105°, 195°, 285°$ **[1 mark]**

6 $2\cos^2(\theta) = 1$

$\cos^2(\theta) = \frac{1}{2}$ **[1 mark]**

$\cos(\theta) = \pm\sqrt{\frac{1}{2}}$

$= \pm\frac{1}{\sqrt{2}}$



[1 mark]

From the triangle, $\cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$

$\theta = \frac{\pi}{4}, \pi - \frac{\pi}{4}, \pi + \frac{\pi}{4}, 2\pi - \frac{\pi}{4}$ (all quadrants) **[2 marks]**

$= \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$ **[1 mark]**

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7 a Amplitude $= 7$ **[1 mark]**

Period $= 4\pi$ **[1 mark]**

b $\frac{2\pi}{b} = 4\pi$

$\frac{2\pi}{4\pi} = b$

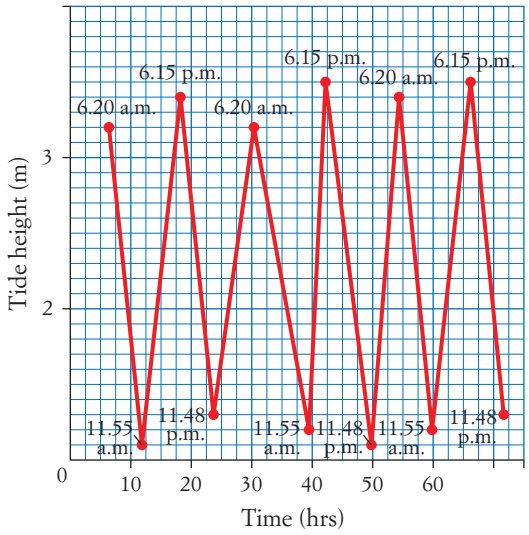
$\frac{1}{2} = b$ **[1 mark]**

Substitute into $y = -a \sin(bx)$ (reflected in the x-axis).

$y = -7 \sin\left(\frac{x}{2}\right)$ **[1 mark]**

Part C

1 a

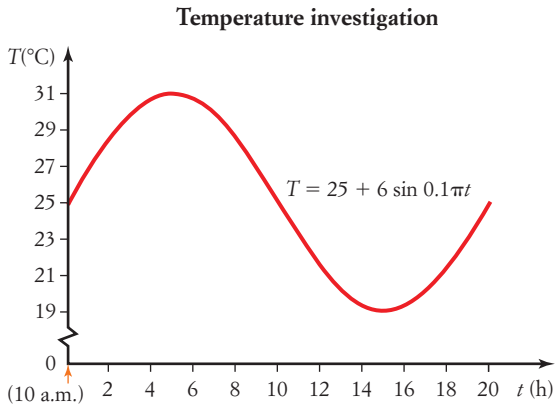


[3 marks]

b Period 24 hours, amplitude 1.25. **[2 mark]**

c About 2.5 m. **[1 mark]**

2 a Period $= \frac{2\pi}{0.1\pi} = 20$, amplitude $= 6$, the graph is translated 25 units upwards. **[1 mark]**



[2 marks]

b From the graph, the highest was 31° C and the lowest was 19° C. **[2 marks]**

c The highest occurred at 3 p.m., the lowest occurred at 1 a.m. **[2 marks]**

d i 11.05 a.m., 6.55 p.m., … **[1 mark]**

ii 11.08 p.m., 2.52 a.m., … **[1 mark]**

e This is unusual because the period is not 24 hours. **[1 mark]**

3 a $\cos(2x) = \cos(x + x)$

$= \cos(x)\cos(x) - \sin(x)\sin(x)$ **[1 mark]**

$= \cos^2(x) - \sin^2(x)$ **[1 mark]**

$= \cos^2(x) - [1 - \cos^2(x)]$ **[1 mark]**

$= \cos^2(x) - 1 + \cos^2(x)$

$= 2\cos^2(x) - 1$ **[1 mark]**

b $\sin(2x) = \sin(x + x)$

$= \sin(x)\cos(x) + \cos(x)\sin(x)$ **[1 mark]**

$= 2\sin(x)\cos(x)$ **[1 mark]**

c $\cos(3x) = \cos(2x + x)$

$= \cos(2x)\cos(x) - \sin(2x)\sin(x)$ **[1 mark]**

$= [2\cos^2(x) - 1]\cos(x) - 2\sin(x)\cos(x)\sin(x)$ **[1 mark]**

$= 2\cos^3(x) - \cos(x) - 2\sin^2(x)\cos(x)$ **[1 mark]**

$= 2\cos^3(x) - \cos(x) - 2[1 - \cos^2(x)]\cos(x)$ **[1 mark]**

$= 2\cos^3(x) - \cos(x) - 2\cos(x) + 2\cos^3(x)$ **[1 mark]**

$= 4\cos^3(x) - 3\cos(x)$ **[1 mark]**

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